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Neural Correlates of Mathematical Problem Solving

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This study explores electroencephalography (EEG) brain dynamics associated with mathematical problem solving. EEG and solution latencies (SLs) were recorded as 11 neurologically healthy volunteers worked on intellectually challenging math puzzles that involved combining four single-digit numbers through basic arithmetic operators (addition, subtraction, division, multiplication) to create an arithmetic expression equaling 24. Estimates of EEG spectral power were computed in three frequency bands — θ (4–7 Hz), α (8–13 Hz) and β (14–30 Hz) — over a widely distributed montage of scalp electrode sites. The magnitude of power estimates was found to change in a linear fashion with SLs — that is, relative to a base of power spectrum, theta power increased with longer SLs, while alpha and beta power tended to decrease. Further, the topographic distribution of spectral fluctuations was characterized by more pronounced asymmetries along the left–right and anterior–posterior axes for solutions that involved a longer search phase. These findings reveal for the first time the topography and dynamics of EEG spectral activities important for sustained solution search during arithmetical problem solving.

Keywords: Mathematical problem solving; electroencephalogram (EEG); solution latencies (SLs).

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1. Introduction

In the current era of information saturation, problem solving — rather than rote knowledge accumulation — has become an increasingly important educational goal.[1] Cognitive neuroscience can offer timely insight into the neurodynamics supporting successful problem solving. Recent work in our labs, for instance, has explored brain activities during engagement in ill-structured problems with open solutions requiring specialized knowledge of optics.[2] Intriguingly, the magnitude of spectral power in a range of frequencies important for cognitive processing — including theta, alpha and beta — was found to increase as solvers came on their own to develop progressively more optimal, elegant solutions. This finding distinguishes problem solving from other domains of human endeavor, such as skill learning, which tends to be characterized by an overall decrease in task-related cortical activity as performance improves with practice.[3,4]

The present study extends our exploration of the relationship between EEG dynamics and performance on intellectually challenging problems. Previously, we uncovered an intriguing relationship between frontal theta activities and solution time.[5] Whereas slow solvers exhibited greater frontal theta over frontal right hemisphere recording sites, the opposite pattern was exhibited by fast solvers. Building upon this discovery, the present study explores this question: How are the magnitude and topographic distribution of EEG spectral dynamics modulated as a function of the amount of time required to solve a problem? On the one hand, problems that are solved quickly and easily might be associated with greater short-term mental effort — and hence, stronger cortical engagement — resulting in speedier performance, whereas problems involving long solution latencies (SLs) might be characterized by relatively diminished measurable spectral activities. On the other hand, however, paralleling our previous work, the opposite pattern could be found — that is, relative to their easier counterparts, difficult problems could lead to greater, and even topographically distinct, forms of engagement. This latter possibility is consistent with the idea that when faced with challenges — whether in the form of mental impasse or other internally generated constraints, or in the form of particularly challenging problem parameters — solvers "redouble their efforts" so to speak, recruiting additional and qualitatively different resources relative to problems that offer more straightforward paths to solution.

To examine these possibilities, we recorded EEG as healthy adults worked on math puzzles derived from a popular commercial game known as Math24. Each problem consisted of four single-digit numbers that were to be combined through addition, subtraction, division, or multiplication to reach a final solution that equaled 24. For example, a puzzle consisting of 1, 7, 1, and 2, could be correctly resolved through the following steps: $1 + 7 = 8$; $2 + 1 = 3$; $3 \times 8 = 24$. The use of math puzzles contrasts significantly with previous studies of problem solving, which have typically involved language-based stimuli, such as anagrams, riddles, and remote associates.[6–8] There are a number of motivations for this approach. First, while mental arithmetic has received extensive attention from a neurocognitive perspective, fairly little is known about problem solving per se in the math domain. Secondly, the abundance of Math24 puzzles — which all involve the same range of numbers (1 through 9), and the same number of steps to solution — affords the opportunity to compare EEG activities over multiple well-controlled trials.

In the present study, participants worked on math puzzles for one hour at their own pace. We expected a wide range of solution times within each individual, allowing for the detailed study of spatio-temporal brain dynamics during both challenging and relatively easy solution searches. Spectral modulations of EEG power were analyzed using event-related spectral perturbation (ERSP) and other common time–frequency-based approaches.[9,10] The relationship between hemispheric and anterior–posterior asymmetries of EEG activation and solution times was also explored. Given previous research indicating that asymmetric patterns of brain activation are related to participants' mathematics ability,[11] we explore the relationship between sustained mental math problem solving and the asymmetric modulation of EEG power over anterior versus posterior recording sites.

2. Methods

2.1. *Subjects*

Eleven healthy adults (nine males, two females, age: 15–49 years) with normal or corrected-to-normal vision participated in the experiment. The

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Institutional Review Board of University of California San Diego approved the experimental protocol used in this study. Subjects were required to abstain from alcoholic, caffeinated or tobacco drinks for 24 h before the recording session. Sessions took place either in the morning (9–12 am) or afternoon (2–5 pm). Participants were instructed regarding the experimental procedures and engaged in practice blocks until they felt confident in their understanding of the task. None of the volunteers was aware of the hypotheses being tested. All participants delivered their informed consent prior to any experimentation.

2.2. *Materials and Procedures*

Math24 is an intellectually challenging game that is commercially available and widely used in schools with the aim of sharpening problem solving and mental math skills (http://www.24game.com/). While a number of variants of the game exist, each puzzle in the present experiment involved four numbers to be combined through basic operations (addition, subtraction, division, multiplication) to reach a final solution equal to 24. For example, a trial with 2, 2, 4, 7, could yield the following solution: $2 + 2 = 4$, $4 * 7 = 28$ and $28 - 4 = 24$. Note that each of the four numbers is used only once and yields three equations as the final solution.

An ideal feature of Math24 puzzles is that difficulty level can be varied while holding all other task parameters constant. Harder problems typically involve fewer possible and more atypical solutions (i.e. they are not readily reduced to a variant of 8×3 or 6×4, and may be solvable by only one approach). In the present study, puzzles representing three levels of difficulty, as ranked by the manufacturer, were presented in a randomly interleaved fashion.

During the experimental session, participants were allotted 120 s to solve each puzzle. They sat in a comfortable chair in front of a 27″ computer monitor, and silently worked on puzzles in a self-paced fashion. Upon reaching a solution, participants were instructed to press a button immediately without taking any time to mentally verify their solutions. Next, they verbally described their solutions to a co-present experimenter seated behind them.

Each trial began with a black background on the computer monitor for 2 s (a baseline period), followed by the numbers that comprised the puzzle, which remained visible on the screen during the entire solution phase (120 s maximum). Problems that were not solved within 120 s "timed-out," and the next trial was presented. Experimental sessions lasted 1 h.

2.3. *Data acquisition*

128-channel EEG data were amplified (BioSemi ActiveTwo EEG system), referenced to CMS-DRL ground, sampled at 512 Hz, and notch filtered at 60 Hz. The contact impedance between the scalp and the electrode was calibrated to be less than 5 kΩ using conductive gel. Electrode locations were digitized using a 3D digitizer (Zebris Medical GmbH).

2.4. *Analysis of behavioral data*

Trials resulting in incorrect solutions or solution times exceeding 120 s were excluded from further analysis. For each correctly solved trial, the solution latency (SL) was computed from the time between stimulus onset and the button press to indicate solution readiness. Each participant's performance was defined as their mean SL, averaging SLs across trials. Repeated-measures one-way ANOVA tests were used to compare groups' mean SLs across the three levels of task difficulty.

2.5. *EEG preprocessing*

MATLAB (The Mathworks, Inc.) and an open source toolbox, EEGLAB (Swartz Center for Computational Neuroscience, La Jolla, CA; http://www.sccn.ucsd.edu/eeglab), were used for all preprocessing and analysis. Figure 1 is a schematic illustration of the analysis pipeline for a single subject. EEG channels were re-referenced offline to an average of the left and right mastoids. A band-pass filter between 2 and 30 Hz was applied to raw, time-series EEG to eliminate line noise (60 Hz and its harmonic) and DC drifts. Subsequently, data were compressed by downsampling to 256 Hz (Fig. 1(A), Step ①) and visually inspected to remove channels and data segments heavily contaminated by nonphysiological artifacts, such as momentary spikes in electrode impedance, or high amplitude voltage fluctuations across all channels introduced by postural shifts (Fig. 1(A), Step ②).

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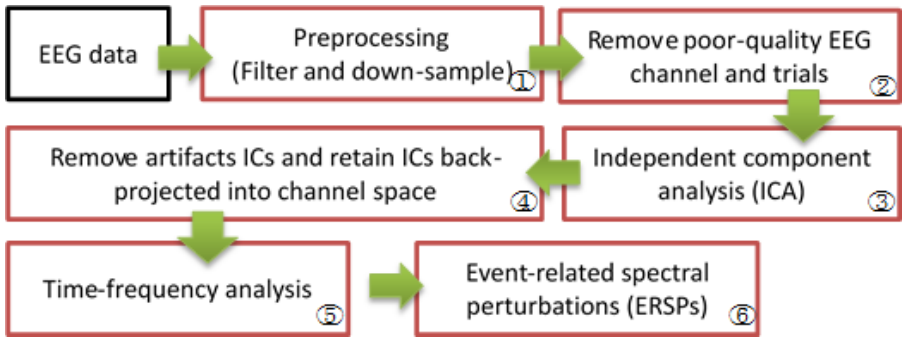
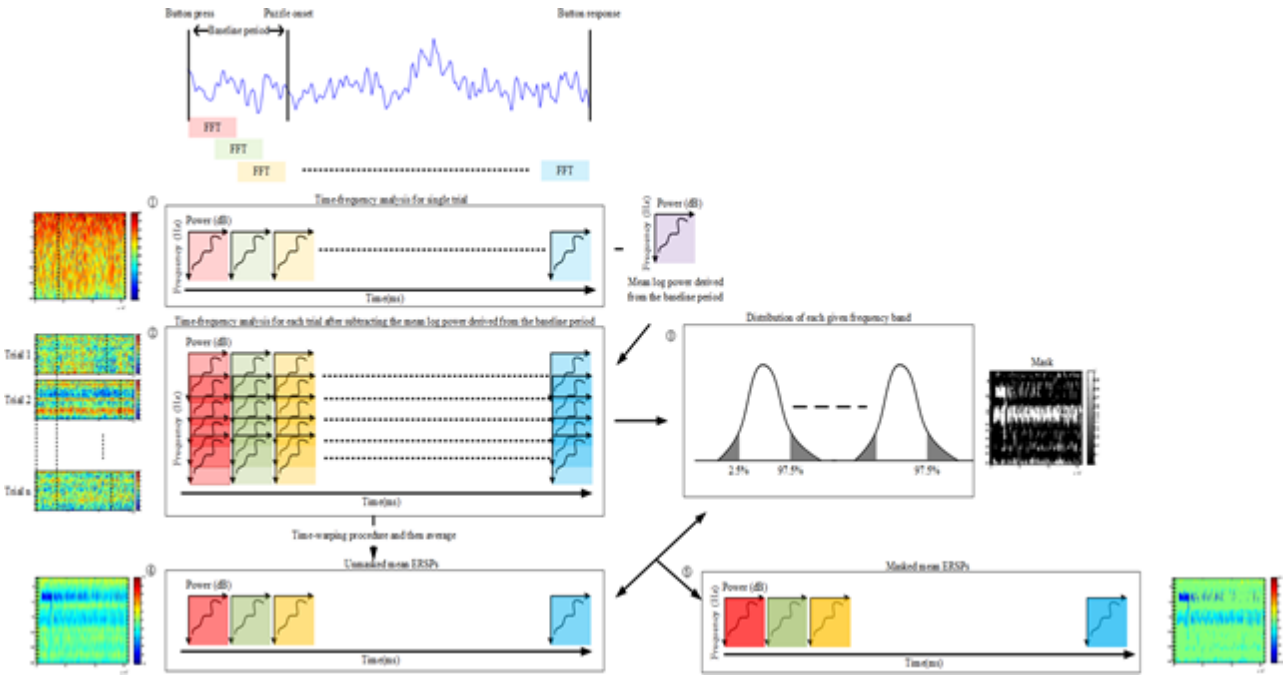


(A)



(B)

Fig. 1. (Color online) (A) Flow chart of EEG analysis steps. Step ①: EEG data preprocessing, including filtering and down-sampling. Step ②: The EEG signals were inspected to remove poor-quality channels and trials heavily contaminated by nonphysiological artifacts. Step ③: ICA was adopted to decompose the recorded EEG signals into temporally independent time courses. Step ④: ICs accounting for blinks, eye moments and muscle activity were visually identified and removed. The remaining ICs were back-projected to channel space. Step ⑤: Time–frequency analysis. Step ⑥: ERSPs. (B) The illustration of procedures for time–frequency and ERSP analysis for a single subject. Step ①: Single-trial data yielding a matrix of frequencies (2–30 Hz) × times (−2–10.3 s) derived by short time Fourier transform (STFT) with Hanning-window sinusoidal wavelets. Step ②: Across single trials, the spectral density estimates were converted to log power, and mean log power derived from the 2-s pre-stimulus baseline period of the same trials was subtracted. Step ③: A distribution of several hundred surrogate values representing random fluctuations of mean spectral power at each of the 114 frequencies. If original mean ERSP values fell outside of the upper or lower extremes of the surrogate distribution, they were considered reliably different from the baseline; otherwise, they were considered nonsignificant and zero-masked. The black and white mask indicates where values fell outside (white) or within (black) the thresholds. Step ④: A linear time-warping procedure was applied to align power spectral density estimates across the time dimension of each trial. Next, the aligned power estimates were averaged across trials. Step ⑤: If a mean ERSP value reliably deviated from baseline, that value was retained in the final plot (more red is positive; more blue is negative). Otherwise the value of the ERSP was plotted as zero (green).

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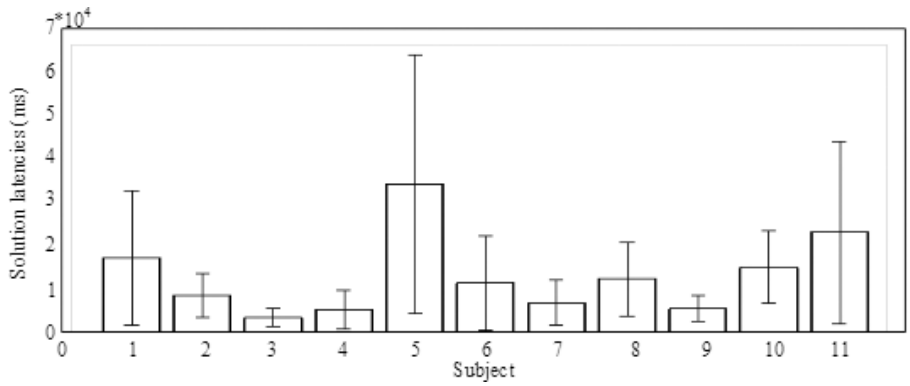
Independent component analysis (ICA) was applied to continuous EEG for the purpose of artifact correction[12–16] (Fig. 1(A), Step ③). ICA is a family of algorithms that learns optimal filters to estimate the activities of spatially coherent cortical sources, as well as nonbrain artifacts, from the statistics of scalp-recorded EEG.[9,17] Resultant independent components (ICs) accounting for numerous kinds of artifacts, including blinks, eye moment, EKG, muscle activity, and line noise can be discerned on the basis of their mean power spectra, time-series activations, and topography of their projections to scalp electrodes. In the artifact correction procedure implemented here, the contributions of such nonbrain sources were removed by zeroing their projection weights to each scalp channel and then re-computing estimations of the original data as a weighted linear sum of the remaining source activities (Fig. 1(A), Step ④).

2.6. *Time–frequency analysis*

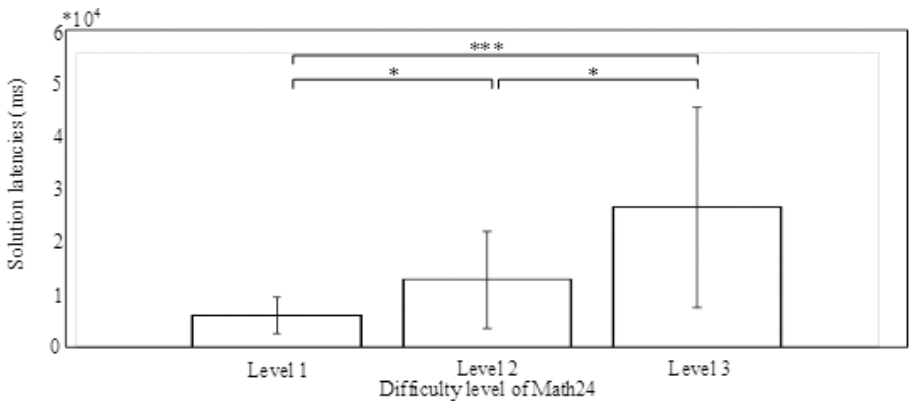
2.6.1. *EEG dynamics during mathematical problem solving*

As illustrated in Fig. 2, mean SLs differed substantially across the three levels of task difficulty. The differences in SLs to easy problems (level 1) across participants were small (mean ± SD: 5.3 ± 4.8 s), whereas on the challenge level (level 3), substantially more variance was observed (mean ± SD: 22.3 ± 30.3 s). Further at level 3, the number of available trials was much fewer than at level 1 and level 2. For this reason, the time–frequency analysis was performed only on intermediate level problems (mean ± SD: 10.3 ± 13.5 s).

For each participant, 23 to 88 (mean ± SD: 47 ± 22) correctly solved, artifact-free trials were available for time–frequency analysis at the intermediate level. Single-trial data epochs were extracted from



(A)



(B)

Fig. 2. (A) Mean SLs to intermediate-level problems (level 2) sorted across subjects. (B) Mean SLs for all 11 subjects on easy (level 1), intermediate (level 2), and difficult (level 3) problems. Statistical test results are noted as $*p < 0.05$, $**p < 0.01$ and $***p < 0.001$.

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artifact-corrected time series EEG within a time window extending from the 2-s pre-stimulus onset to the button press indicating solution readiness. Although EEG was recorded via a 128-channel Quik-cap, preliminary analysis revealed that neighboring channels yielded comparable results — likely as a result of volume conductance, in keeping with other work using a similar paradigm.[18,19] Thus, we selected 12 representative recording sites (Frontal: F3, F4; Central: C3, C4; Parietal: P3, P4; Occipital: O3, O4; Midline: Fz, Cz, Pz and Oz) for further analysis.

Single-trial data from each of these sites were convolved by STFT with Hanning-window sinusoidal wavelets[9] spanning a frequency range of 2 to 30 Hz. To visualize EEG spectral dynamics over the entire course of the problem solving period, estimates of spectral power were computed across 200 equally spaced time points, yielding a matrix of frequencies (2–30 Hz) × times (−2–10.3 s) (Fig. 1(A), Step ⑤ and Fig. 1(B), Step ①). For each subject, ERSPs were derived by converting spectral density estimates to log power and subtracting the mean log power derived from the 2-s pre-stimulus baseline period (Fig. 1(A), Step ⑥ and Fig. 1(B), Steps ② and ④). Because SLs varied widely across trials and subjects, a linear time-warping procedure was applied to align power spectral density estimates across the time dimension of each trial with the median latency of all trials (7 s), yielding an overall time window of 9 s (including the prestimulus baseline). Then, the aligned power spectral density estimates were averaged across trials. For a detailed description of the time-warping procedure, please see the online tutorial at http://www.sccn.ucsd.edu/eeglab.

A nonparametric bootstrap method was used to test the reliability of mean ERSPs obtained for each channel and participant. A distribution of several hundred surrogate values representing random fluctuations of mean spectral power at each of the 114 frequencies was generated by randomly and repetitively shuffling values across the 200 time points within each given frequency band of each trial and then averaging (Fig. 1(B), Step ③). Actual obtained values within each frequency band were then compared with the upper and lower extremes of the surrogate distribution (2.5% and 97.5% percentile for a two-tailed test where $\alpha = 0.05$). If original mean ERSP values fell outside of this envelope, they were considered reliably different from the baseline; otherwise, they were considered nonsignificant and zero-masked (Fig. 1(B), Step ⑤). For each channel, grand average ERSPs were computed by averaging across all 11 participants' significance-masked mean ERSPs. Statistical reliability of grand averages was assessed in a manner similar to that employed in the analyses of individual participants. Time–frequency and ERSP analyses can be accomplished by other methods (e.g. continuous wavelet transform (CWT) plots[20] and Morlet wavelet transforms followed by exploratory parallel-factor analysis[21]) and will be considered in the future work. Due to their high resolution, both of these methods are, particularly useful for the analysis of transients, aperiodicity, and other nonstationary signal features of clinical significance. However, the STFT-based approach implemented here is also well established.[9]

2.6.2. *Performance-sorted single trial EEG spectra*

To investigate the relationship between EEG spectral activities and solution times across individuals, the mean power spectrum (2–30 Hz) was computed for trials of intermediate difficulty as follows: estimates of spectral power for each frequency step were derived from puzzle onset to the button press signaling solution readiness (Fig. 1(B), Step ①). Mean power spectra of each trial were obtained by averaging estimates across the time dimension and then subtracting the base of power spectrum, which was comprised of mean power spectral densities averaged across trials in which participants responded very quickly (within 5 s or fewer). Mean power spectra were pooled across subjects and sorted by their SLs. Subsequently, the performance-sorted EEG spectra were smoothed through an equal-weight moving average method,[22] with a window of 9 s and a step of 0.5 s.

2.6.3. *Topographic distribution of EEG spectral modulations as a function of task performance*

To assess the relationship between task performance and the topographic distribution of EEG spectral activities, all trials of intermediate difficulty were divided into three groups according to SL — Group I (SLs $\leq$ 20 s, 31 trials), Group II (20 s $<$ SLs $\leq$ 40 s, 40 trials) and Group III (SLs $>$ 40 s, 40 trials).

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This study compared and contrasted the averaged power spectra across these three groups of trials. Within each group, single-trial power spectra were computed according to the same method described in Sec. 2.6.2, but without subtracting the base of power spectrum because it belonged to the Group I. Estimates of θ (4–7 Hz), α (8–13 Hz) and β (14–30 Hz) power spectral density were derived from stimulus onset to the button press indicating solution readiness. Spectral modulations within these fixed frequency bandwidths have been robustly implicated in cognitive processing.[23–25] To assess the topographical distribution of EEG power across the three rates of problem solving, two types of subtractions (laterality and caudality) were performed using the mean of trial estimates of theta, alpha and beta band power within each group. Laterality measures of band power associated with right hemisphere electrodes were subtracted from their left hemisphere counterparts, and caudality measures associated with electrodes over the back of the head (P3/Pz/P4) were subtracted from those obtained from anterior electrodes (F3/Fz/F4). Paired-sample repeated-measure ANOVA tests were used to assess the reliability of EEG power differences detected in cerebral laterality and caudality across the three groups.

3. Results

3.1. *Accuracy and SLs*

The quantity of mental math problems solved within 1 h varied widely ($n = 66$–188). Given the relatively small number of participants in the present study, it was not feasible to compare EEG activities in fast and slow solvers. Figure 2(A) shows the distribution of SLs at level 2 across all individual participants, while Fig. 2(B) shows the mean effect of

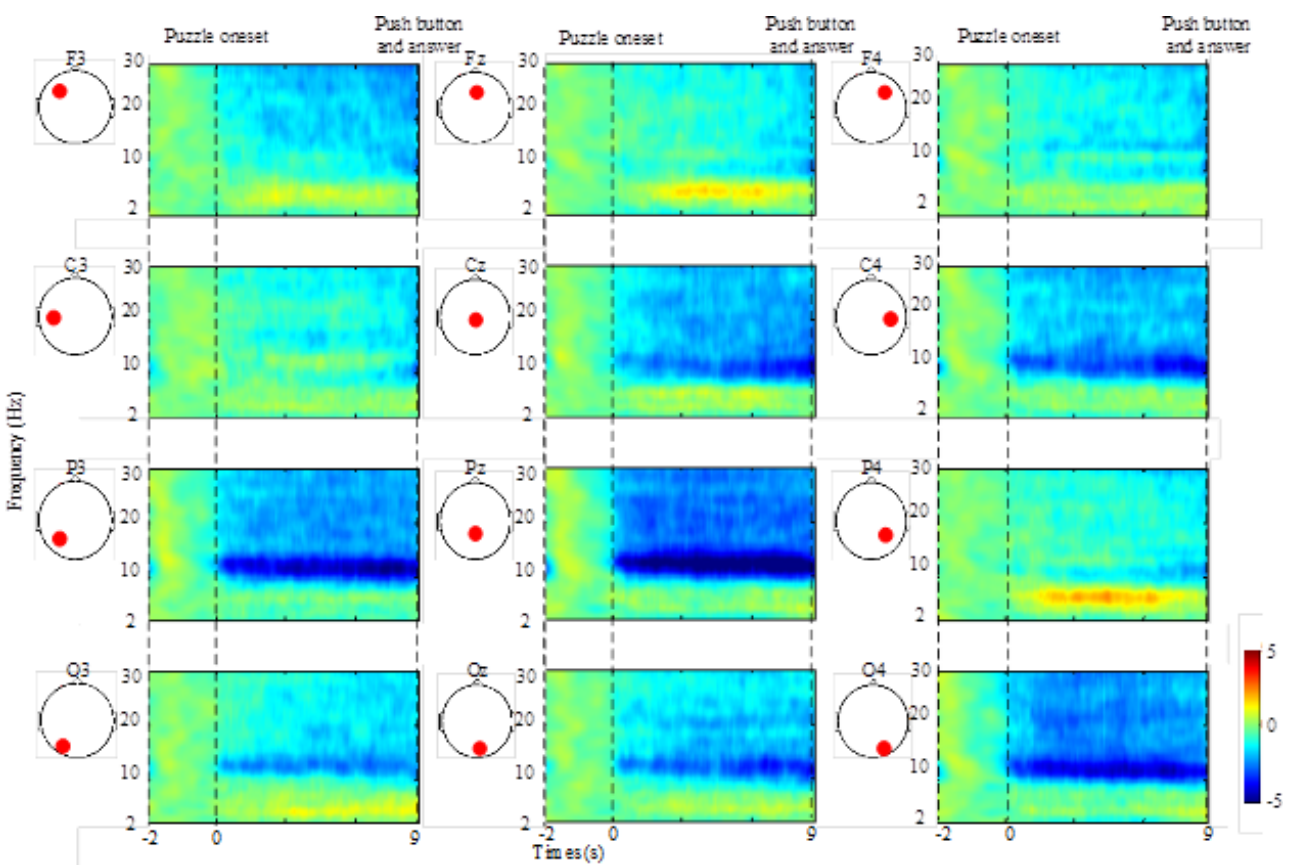

Fig. 3. (Color online) Event-related spectral activities obtained from 12 scalp recording sites over a time interval extending from the onset of puzzles to the button press indicating solution readiness. Power (in dB) is color coded with increases up to 5 dB in red and decreases down to −5 dB in blue relative to the pre-stimulus baseline. Time is plotted on the x-axis in 1000 ms increments. Frequencies (2 to 30 Hz) are plotted on the y-axis.

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task difficulty across the three levels. A reliable main effect of difficulty level ($F(2, 30) = 7.78$, $p < 0.001$) warranted follow-up contrasts. As expected, level one puzzles were solved more rapidly than those of intermediate difficulty ($F(1, 20) = 5.18$, $p < 0.05$), and intermediate problems, in turn, were solved more quickly than level three items ($F(1, 20) = 4.6$, $p < 0.05$).

3.2. *EEG dynamics during mathematical problem solving*

Figure 3 shows ERSPs during the problem-solving phase from puzzle onset to solution readiness at 12 representative scalp locations for intermediate-level problems. The vertical lines at the right edge of each plot represent the button press, indicating readiness to give a solution. A broadly distributed, sustained suppression of activities in the alpha and beta ranges (10–30 Hz) is visible throughout most of the epochs over Pz and P3, and to a lesser degree over Cz, C4 and occipital recording sites. Notably, over other channels — particularly those characterized by event-related synchronization (ERS) in the theta range — a pattern of progressively increasing alpha and beta suppression can be discerned, with maximal desynchronization in the final 3 s of the problem solving phase. Finally, prominent theta ERS occurs over the frontal midline around two and a half seconds post-stimulus and over P4 around 1 s post-stimulus. In both cases, theta power declines approximately 3 s before a solution is reached.

3.3. *Performance-sorted EEG spectra*

Figure 4 shows single-trial EEG power spectra pooled across participants and sorted by SL. Notably, theta power over Fz and P4 increased with increasing SLs. Further, EEG power over F3, F4 and P4 increased in amplitude in the beta-band for SLs

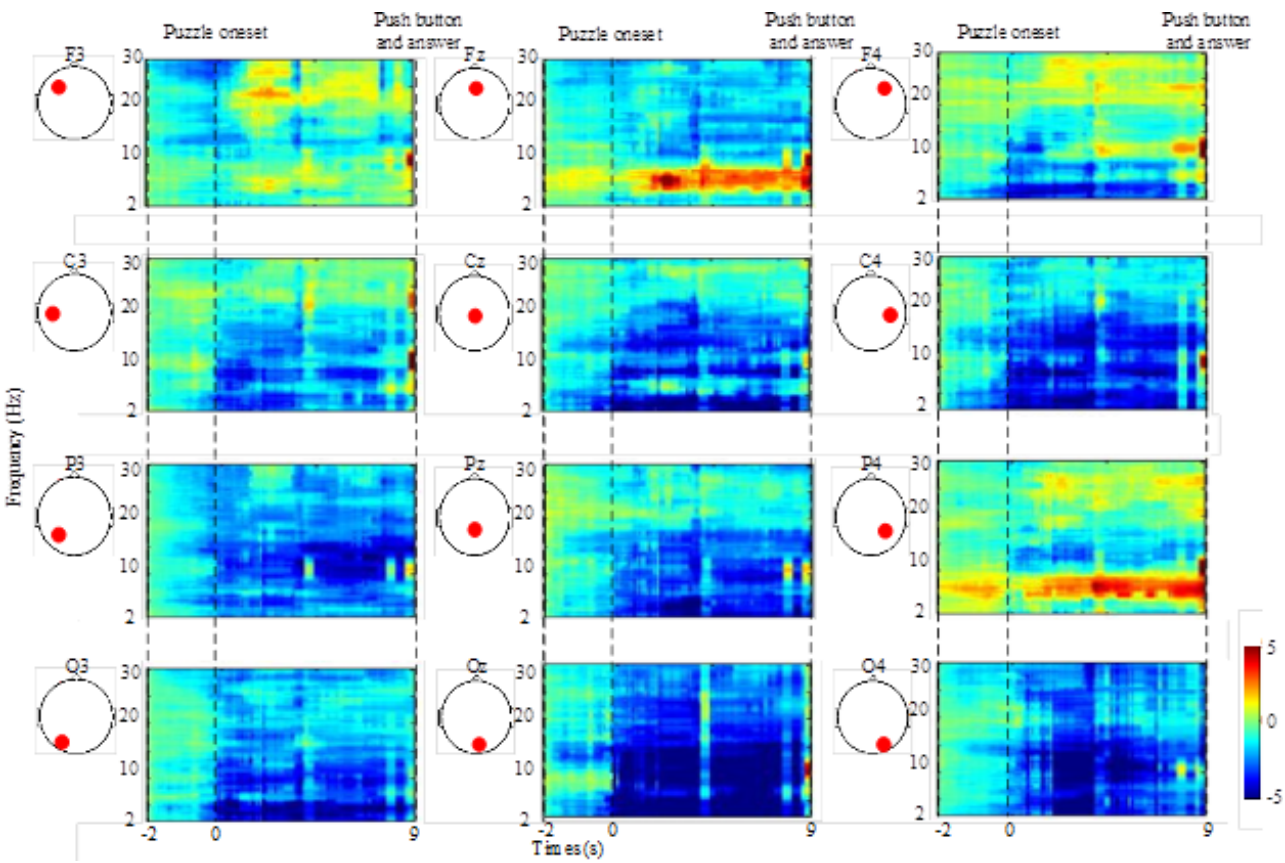

Fig. 4. (Color online) Single trial power spectra sorted by solution times. Power (in dB) is color coded with increases up to 5 dB in red and decreases down to −5 dB in blue relative to the pre-stimulus baseline. Solution times are plotted on the x-axis in 1 s increments. Frequencies (2 to 30 Hz) are plotted on the y-axis (high to low).

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longer than 37 s. A decrease in the power across all of the frequency bands (2–30 Hz) of EEG activity was observed at F4, C3, Cz, C4, P3, Pz, O3, Oz, O4 when solutions required 20 s or more.

3.4. *Hemispheric and anterior–posterior asymmetries in EEG power are linked to solution times*

Table 1 and Fig. 5 show EEG power differences derived by subtracting estimates associated with RH recording sites from their counterparts over the LH in three groups of trials created on the basis of solution times. For trials characterized by intermediate solution times (Group II), beta-band activities were greater over F3 (LH) than F4 (RH), whereas the opposite was true for trials in Group I — that is, beta power estimates obtained over F3 were lower than over F4. No reliable hemispheric differences were detected in association with central electrodes. However, over parietal channels (P3 versus P4), greater right versus left hemisphere theta power was found for all three groups of trials. This asymmetry was

Table 1. Summary of statistics for EEG power differences derived by subtracting power estimates associated with RH recording sites from their counterparts over the LH in three aggregates of trials grouped by solution times.

Cerebral laterality	Frequency (Hz)	Group	Difference (dB)	Degrees of freedom	F-statistic	p-value
F3 versus F4	θ	I	-0.30 ± 0.35	F(1,62)	2.57	$p = 0.11$
		II	1.49 ± 0.50	F(1,78)	37.61	$p < 0.001$
		III	0.20 ± 0.65	F(1,80)	1.54	$p = 0.21$
	α	I	-1.04 ± 0.21	F(1,62)	75.02	$p < 0.001$
		II	0.17 ± 0.51	F(1,78)	0.54	$p = 0.46$
		III	-0.75 ± 0.48	F(1,80)	14.96	$p < 0.001$
	β	I	-0.88 ± 0.42	F(1,62)	78.12	$p < 0.001$
		II	0.88 ± 1.11	F(1,78)	16.78	$p < 0.001$
		III	-0.14 ± 0.53	F(1,80)	4.22	$p < 0.05$
C3 versus C4	θ	I	-0.11 ± 0.37	F(1,62)	0.20	$p = 0.64$
		II	-0.27 ± 0.38	F(1,78)	1.59	$p = 0.21$
		III	0.06 ± 0.29	F(1,80)	0.10	$p = 0.75$
	α	I	-0.03 ± 0.35	F(1,62)	0.01	$p = 0.90$
		II	-0.42 ± 0.59	F(1,78)	2.58	$p = 0.11$
		III	-0.57 ± 0.61	F(1,80)	3.30	$p = 0.07$
	β	I	-0.03 ± 0.36	F(1,62)	0.060	$p = 0.79$
		II	0.23 ± 0.32	F(1,78)	2.56	$p = 0.11$
		III	0.60 ± 0.31	F(1,80)	47.05	$p < 0.001$
P3 versus P4	θ	I	-2.14 ± 0.74	F(1,62)	194.39	$p < 0.001$
		II	-1.18 ± 1.86	F(1,78)	13.24	$p < 0.001$
		III	-4.57 ± 0.53	F(1,80)	1260.78	$p < 0.001$
	α	I	0.40 ± 0.29	F(1,62)	15.54	$p < 0.001$
		II	1.80 ± 1.39	F(1,78)	41.83	$p < 0.001$
		III	-0.65 ± 0.63	F(1,80)	8.42	$p < 0.01$
	β	I	-0.82 ± 0.34	F(1,62)	58.24	$p < 0.001$
		II	0.11 ± 1.37	F(1,78)	0.21	$p = 0.64$
		III	-1.95 ± 0.4	F(1,80)	557.69	$p < 0.001$
O3 versus O4	θ	I	0.09 ± 0.53	F(1,62)	0.24	$p = 0.62$
		II	0.09 ± 0.8	F(1,78)	0.22	$p = 0.64$
		III	0.92 ± 0.64	F(1,80)	29.28	$p < 0.001$
	α	I	-1.10 ± 0.39	F(1,62)	49.10	$p < 0.001$
		II	-0.43 ± 1.11	F(1,78)	4.10	$p < 0.05$
		III	-0.31 ± 0.98	F(1,80)	1.75	$p = 0.18$
	β	I	-0.27 ± 0.29	F(1,62)	9.67	$p < 0.01$
		II	-0.51 ± 0.72	F(1,78)	11.59	$p < 0.001$
		III	0.16 ± 0.55	F(1,80)	18.26	$p < 0.001$

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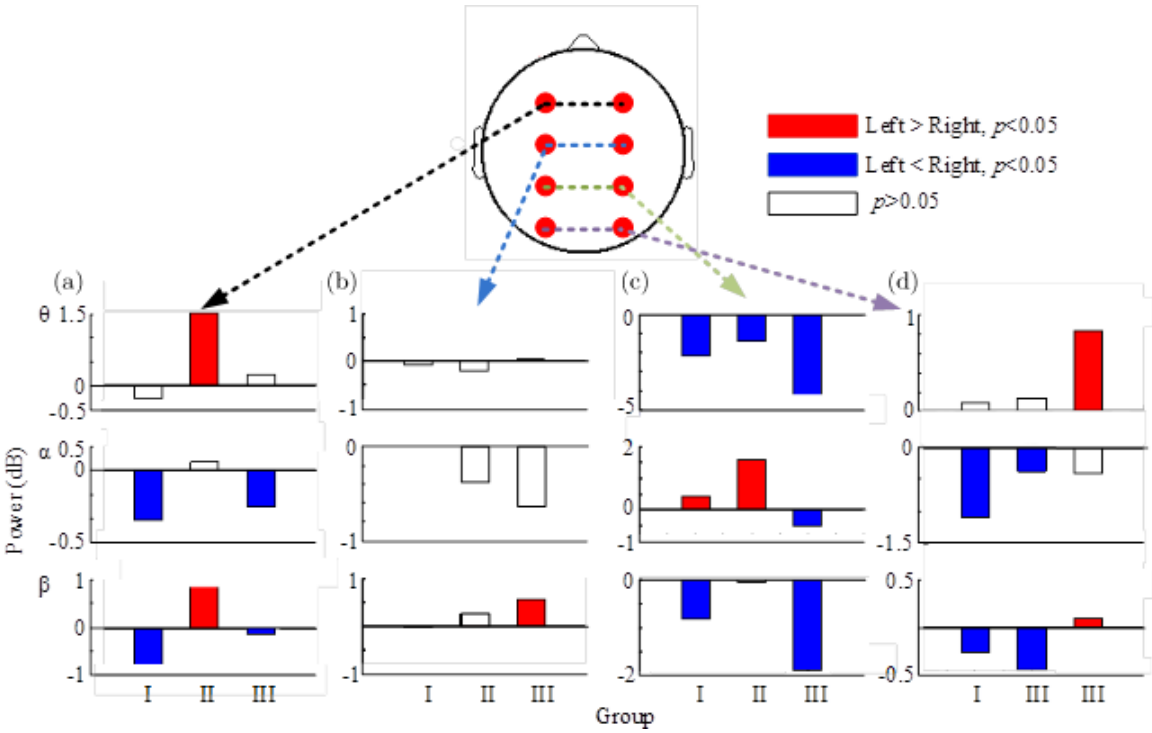

Fig. 5. (Color online) The relationship between hemispheric asymmetry in EEG power modulations and task performance. (A) F4 spectral density estimates subtracted from F3. (B) C4 spectral density estimates subtracted from C3. (C) P4 spectral density estimates subtracted from P3. (D) O4 spectral density estimates subtracted from O3. In Group I, puzzles were solved within 20 s; in Group II, solution times ranged between 20 and 40 s; in Group III, they ranged between 40 s and 2 min. Blue bars indicate reliably greater power over right versus left hemisphere electrodes, whereas red bars represent the opposite.

strongest for the problems requiring the greatest length of time to solve. Alpha-band power exhibited a similar RH dominance for Group III trials as well, whereas LH alpha dominance was found for trials requiring less time to solution (Groups I and II). Finally, over occipital channels, hemispheric asymmetries in beta band power were discovered. Specifically, Group I and II trials led to stronger beta activities over the right versus left hemisphere, while Group III trials resulted in LH beta dominance.

Table 2 and Fig. 6 illustrate the subtraction of EEG power estimates obtained over posterior electrodes from those obtained over anterior ones in the three different groups of trials. For the left

Table 2. Summary of statistics for the anterior minus posterior subtraction of EEG power in the three different groups of trials.

Cerebral laterality	Frequency (Hz)	Group	Difference (dB)	Degrees of freedom	F-statistic	p-value
F3 versus P3	θ	I	0.60 ± 0.37	F(1,62)	16.77	$p < 0.001$
		II	1.13 ± 0.82	F(1,78)	27.43	$p < 0.001$
		III	2.4 ± 0.34	F(1,80)	430.76	$p < 0.001$
	α	I	-0.45 ± 0.22	F(1,62)	17.91	$p < 0.001$
		II	0.030 ± 0.82	F(1,78)	0.020	$p = 0.86$
		III	1.86 ± 0.40	F(1,80)	84.02	$p < 0.001$
	β	I	0.28 ± 0.13	F(1,62)	4.61	$p < 0.05$
		II	1.4 ± 0.51	F(1,78)	32.41	$p < 0.001$
		III	2.02 ± 0.22	F(1,80)	350.32	$p < 0.001$

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Table 2. (*Continued*)

Cerebral laterality	Frequency (Hz)	Group	Difference (dB)	Degrees of freedom	*F*-statistic	*p*-value
Fz versus Cz	θ	I	-3.21 ± 0.50	F(1,62)	742.2	$p < 0.001$
		II	-1.05 ± 0.98	F(1,78)	14.62	$p < 0.001$
		III	0.070 ± 0.47	F(1,80)	0.35	$p = 0.55$
	α	I	-4.57 ± 0.49	F(1,62)	1870.96	$p < 0.001$
		II	-3.4 ± 0.50	F(1,78)	188.12	$p < 0.001$
		III	-3.03 ± 0.35	F(1,80)	192.36	$p < 0.001$
	β	I	-3.33 ± 0.27	F(1,62)	890.99	$p < 0.001$
		II	-2.1 ± 0.98	F(1,78)	86.22	$p < 0.001$
		III	-2.78 ± 0.29	F(1,80)	1243.72	$p < 0.001$
P4 versus P4	θ	I	-1.23 ± 0.58	F(1,62)	42.23	$p < 0.001$
		II	-1.53 ± 1.03	F(1,78)	20.16	$p < 0.001$
		III	-2.37 ± 0.53	F(1,80)	187.65	$p < 0.001$
	α	I	0.99 ± 0.27	F(1,62)	72.1	$p < 0.001$
		II	1.65 ± 0.63	F(1,78)	28.1	$p < 0.001$
		III	1.96 ± 0.30	F(1,80)	86.74	$p < 0.001$
	β	I	0.34 ± 0.13	F(1,62)	34.24	$p < 0.001$
		II	0.63 ± 0.51	F(1,78)	8.23	$p < 0.01$
		III	0.21 ± 0.21	F(1,80)	16.03	$p < 0.001$

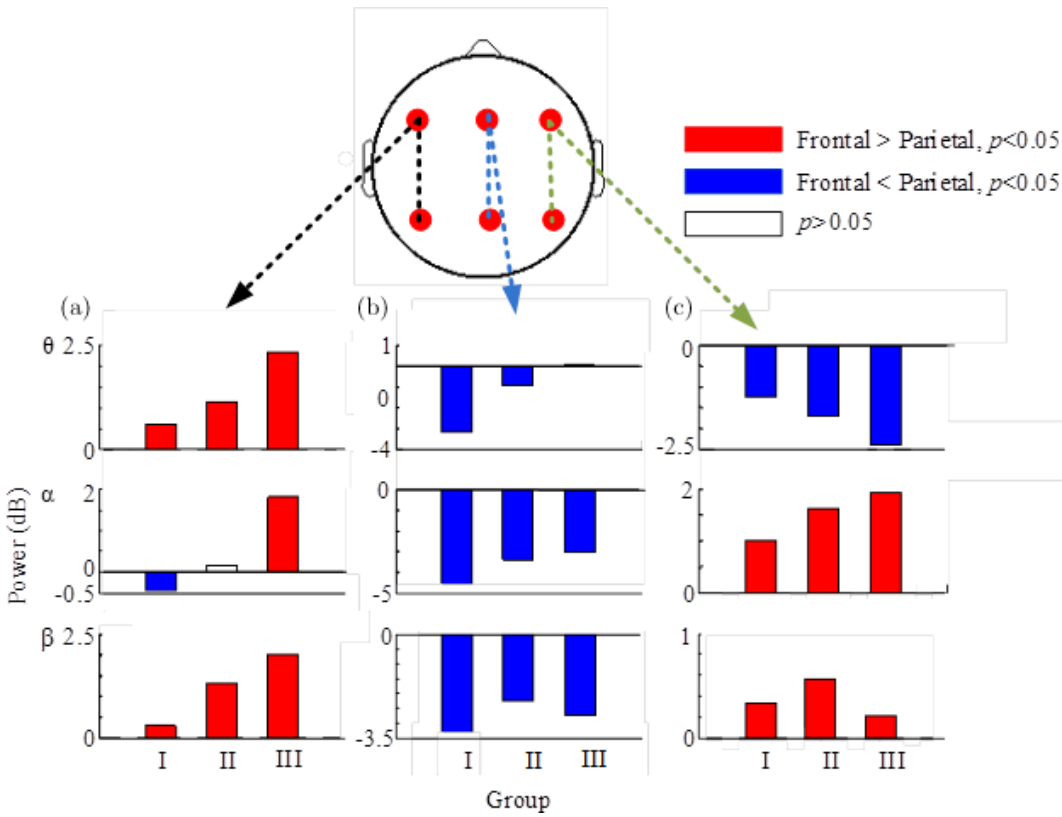

Fig. 6. (Color online) The anterior–posterior distribution of EEG power correlates with task performance. (A) P3 spectral density estimates subtracted from F3. (B) Pz spectral density estimates subtracted from Fz. (C) P4 spectral density estimates subtracted from F4. Blue bars indicate reliably greater power over the back versus the front of the head; red bars indicate the opposite.

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hemisphere (F3 minus P3), all groups exhibited greater theta and beta activities over the front versus the back of the head. This asymmetry grew more pronounced with increasing solution times. For midline electrodes (Fz minus Pz), the opposite pattern obtained — that is, there were greater alpha and beta activities over Pz than Fz. These power asymmetries were greatest for problems solved within 20 s, and decreased in magnitude as solution times increased. For the right hemisphere (F4 minus P4), alpha and beta power were greater over the front versus the back of the head, whereas the opposite pattern occurred in the theta bandwidth. The magnitude of theta and alpha power asymmetries increased linearly with trial group — that is, the asymmetries were enhanced as solution times grew longer.

4. Discussion

Problem solving continues to be of high importance in education and a pivotal ability for students.[26–28] Quantitative problem solving has been a long withstanding difficulty faced by students and has received extensive attention in the field of mathematical development and education.[29] In this study, EEG was recorded as healthy adults solved mental math problems with three levels of difficulty. ERSP, a time–frequency analytical method, were used to evaluate averaged dynamic changes of the broadband EEG spectrum from the onset of math problems to the button press indicating solution readiness. ERSP images revealed alpha- and beta-power suppression and theta-power augmentation during the course of problem solving, consistent with previous EEG studies in mental calculation and working memory (WM) tasks.[30–32] However, the present analysis extends existing knowledge by offering new insight into the detailed spatio-temporal dynamics of spectral changes during problem solving, indicating the engagement of a broadly distributed fronto-parietal brain network whose activities exhibit overlapping, but discrete time courses.

Two topographically and temporally distinct patterns of alpha and beta event-related desynchronization (ERD) were discovered. First, over the posterior regions, ERSP plots revealed tonic, sustained suppression of activities in the alpha and beta ranges (10 to 30 Hz) from puzzle onset to completion. Because it is typically detected over areas of task relevant cortex during active engagement, alpha ERD is generally thought to reflect a release of cortical inhibition,[33] and has been found to increase in magnitude when parameters of task difficulty, such as WM load, are made more challenging.[34] We propose that the sustained ERD in the alpha and beta ranges reflects background processes — likely linked to attentional control — supporting the solution search. This idea is consistent with existing research revealing modulations of alpha and beta ERD[35] as a function of attentional load.

While continuous cortical engagement from select brain regions is clearly necessary for this task, it is also the case that some processes make graded or incremental contributions, as evidenced by the suppression of power above 8 Hz over primarily anterior electrodes (but also P4). In these cases, ERD is typically most apparent in the final few seconds of the problem-solving phase, and weakly discernible at earlier intervals. This pattern suggests that certain cortical systems become increasingly active as a solution is about to be reached.

ERS in the theta range was found predominantly over the frontal midline (Fz) and right parietal (Pz) channels. This pattern of spectral activity is consistent with a number of studies reporting either the engagement of a broadly distributed fronto-parietal theta network[36–38] or primarily frontal theta power augmentation[39,40] during mental arithmetic. It is also consistent with neuroimaging work reporting activation of frontal and parietal cortices, including the right and left inferior frontal gyri, along with the right and left angular gyri and intraparietal sulci, by mental math tasks.[41,42]

Notably, frontal and frontal midline theta has been extensively implicated in tasks requiring attentional control, such as error monitoring[43] or maintenance of verbal and visuo-spatial information in immediate memory.[34,44–52] Whereas the frontal midline theta ERS observed in this study is likely to reflect the engagement of attention-related, domain general executive function, the parietal ERS may reflect more task-specific processes. For instance, Mizuhara and Yamaguchi[37] report long-range theta phase synchronization between clusters of frontal midline and left and right parietal recording sites — and on the basis of simultaneously recorded functional magnetic resonance imaging (fMRI), the authors propose that theta oscillations dynamically

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link frontal brain regions implicated in central executive or other controlled processing with more distal cortical regions, such as the intraparietal sulcus or angular gyrus, which are important for quantitative reasoning.[42,53]

A novel and important contribution of the present work is the snapshot that it affords of the time course of theta activities during solution search. In particular, right parietal theta ERS is visible by approximately 1 s after puzzles appear on the screen and precedes the onset of frontal midline theta ERS by about 1.5 s. However, over both recording sites, discernable power augmentation declines approximately 3 s before participants signal solution readiness. In other words, whereas continuous frontal-parietal theta activities characterize a substantial portion of the problem-solving phase, they are not engaged during the final moments when presumably, participants are formulating a solution. Rather, in the final seconds, ERD in the alpha and beta ranges is the predominant form of spectral modulation.

Overall, on the basis of ERSP outcomes, we propose a three-stage model of problem solving. An initial orienting phase is characterized primarily by tonic alpha and beta ERD visible over most of the scalp and sustained throughout the trial. A subsequent stage that likely reflects exploration of the solution space involves fronto-parietal theta ERS concurrently with the tonic, broadband alpha and beta ERD. Lastly, in the final few seconds before the button press, theta ERS attenuates, while alpha and beta ERD become stronger over electrode sites where they had previously been only weakly detectable. Future work using individually derived, rather than fixed, frequency windows[54] may allow for more fine-grained examination of these processes.

A second important finding of the present work is the positive relationship between the magnitude of theta ERS and SL. As can be seen in Fig. 4, trials involving latencies shorter than 20 s tended to involve less theta ERS over Fz and P4 than those incurring longer latencies. This outcome parallels the finding that mean theta power increases parametrically with increasing WM load,[48,50] and is consistent with the idea that the magnitude of oscillatory activities in the theta range reflects mental effort. Intriguingly, in the present case, difficult problems are characterized not only by longer solution searches, but also increased engagement of mental resources during those search periods.

Our previous work[5] has revealed that fast problem solvers exhibited greater left frontal theta activity, whereas slow solvers showed greater right frontal theta activity. In other words, depending upon degree of math puzzle proficiency, solvers appear to have engaged frontal cortical systems differently. One interpretation of this topographic difference is that the left versus right hemispheric dominance may reflect preferences for distinct problem solving styles. Research on insight problems, for instance, has implicated the right hemisphere in diffuse attention,[55] unconscious processing,[56] and broad semantic representation.[57,58] These aspects of cognition may contribute to the restructuring necessary to connect problem elements in such a way that insight can be suddenly achieved. In the present case, it is possible that slow solvers may have relied more on insight-based approaches than fast solvers, who likely developed efficient search-based strategies.

Support for this idea can be found in the analysis by trials revealing that right parietal theta activities were greater than those over the contralateral left recording site, particularly for trials involving the longest solution times (Fig. 5). In other words, irrespective of subject grouping, both slow and fast solvers tended to recruit the right hemisphere more than the left when problems proved difficult and engendered extended solution searches. While it is not known whether solutions on these trials were ultimately derived through insight, we have recently demonstrated that when participants subjectively rate their experience of insight on each puzzle, high insight ratings tended to be associated with longer solution times.[19]

The analyses by trials also revealed a strong relationship between fronto-parietal asymmetry and task performance (Fig. 6). Fronto-parietal asymmetries in the theta and beta range over LH electrodes, the alpha range over midline electrodes, and theta and alpha range over RH sites all increased linearly across the three groups of trials — that is, with longer solution times the magnitude of the asymmetry became more pronounced. For contrasts involving left and right lateralized recording sites, spectral activities were greater over the front versus the back of the head, whereas the opposite was found for midline channels. An important conclusion that we can infer

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from this pattern of outcomes is that fronto-parietal networks are increasingly engaged as greater obstacles to problem resolution (reflected in solution time) are experienced.

This idea is consistent with previous research implicating fronto-parietal connectivity and activation in the development of abstract numerical concepts,[59–61] in mathematical processing,[11] and problem solving in children and adults.[62] In particular, the right fronto-parietal asymmetry observed here (Fig. 6) coincides with neuroimaging studies demonstrating an analogous right lateralized asymmetry during the mental manipulation of numbers and orienting of attention along the mental number line.[41] Further, in keeping with Micheloyannis *et al.*,[63] who report that neuronal populations in both hemispheres become increasingly engaged during complex arithmetic, we found that when longer amounts of time were required to solve Math24 puzzles, widespread desynchronization occurred within fronto-parietal systems (Fig. 4) — likely due to the prolonged cognitive demands associated with maintaining information in WM, retrieving learned arithmetic facts (e.g. the product of 4×4), and searching for the correct solution path (e.g. discovering that a key to a given problem lies in combining the integers to reach a final equation of 6×4, as opposed to other possible approaches, such as $48 \div 2$).

Finally, the present study supports and extends our own earlier work. We previously reported that within participants, progressively more optimal solutions to open-ended physics problems are characterized by a broadband increase in spectral power across the theta, alpha and beta ranges.[2] Analogously, here we find that difficult solutions are associated with increased theta activities — and intriguingly, with an increasingly asymmetric distribution of spectral activity along the anterior–posterior dimension. In other words, difficult problems elicit not only greater cortical engagement — but also topographically distinct brain areas differently. This finding is particularly important with respect to the prospect of monitoring learners in real time through wearable biosensors, as it suggests that the topography of EEG activities can differentiate brain states associated with difficult and easy problem solving. Future work centered on the co-modulation of EEG spectral power or coherence may yield further insight into underlying mechanisms mediating this effect.

5. Conclusion

Problem solving is an increasingly important educational goal. This study expands upon previous research examining brain processes involved in arithmetic-based problem solving. Over the full time course of the problem-solving phase (from puzzle onset to resolution), spectral modulations of EEG were characterized by augmentation of power in the theta range over the front of the head and P4, as well as a broadly distributed suppression of power in the alpha and beta ranges. These results are consistent with existing research suggesting a fronto-parietal cortical network supporting arithmetic abilities. Further, our work offers the novel finding that longer solution times were linked to the magnitude of spectral modulation. Additionally, the length of solution times was associated with increasingly asymmetric broadband activities along the fronto-parietal axis. This work is important for understanding the neurodynamic foundation of both mathematical cognition and problem solving.

Acknowledgments

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